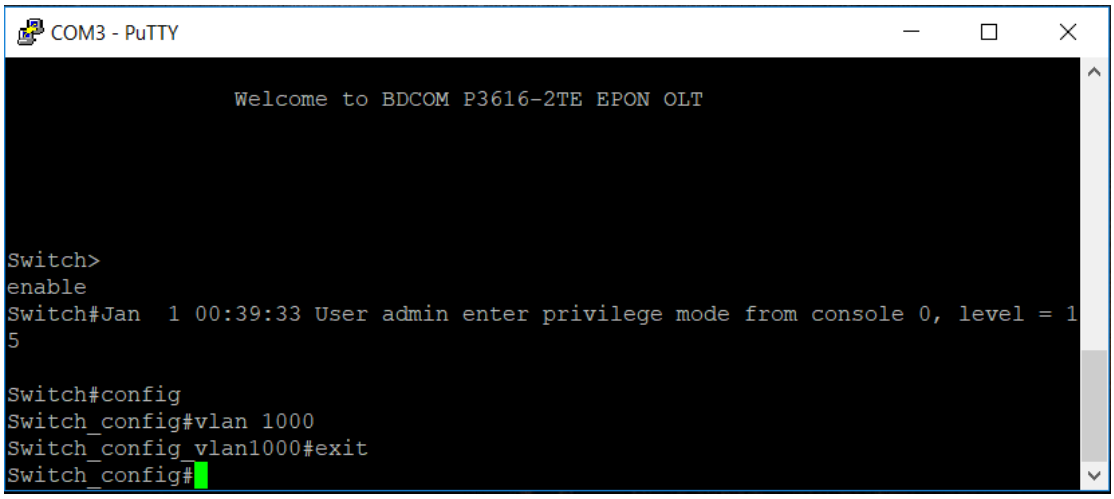
Configuring Multicast in EPON and Testing

To create a VLAN in OLT:

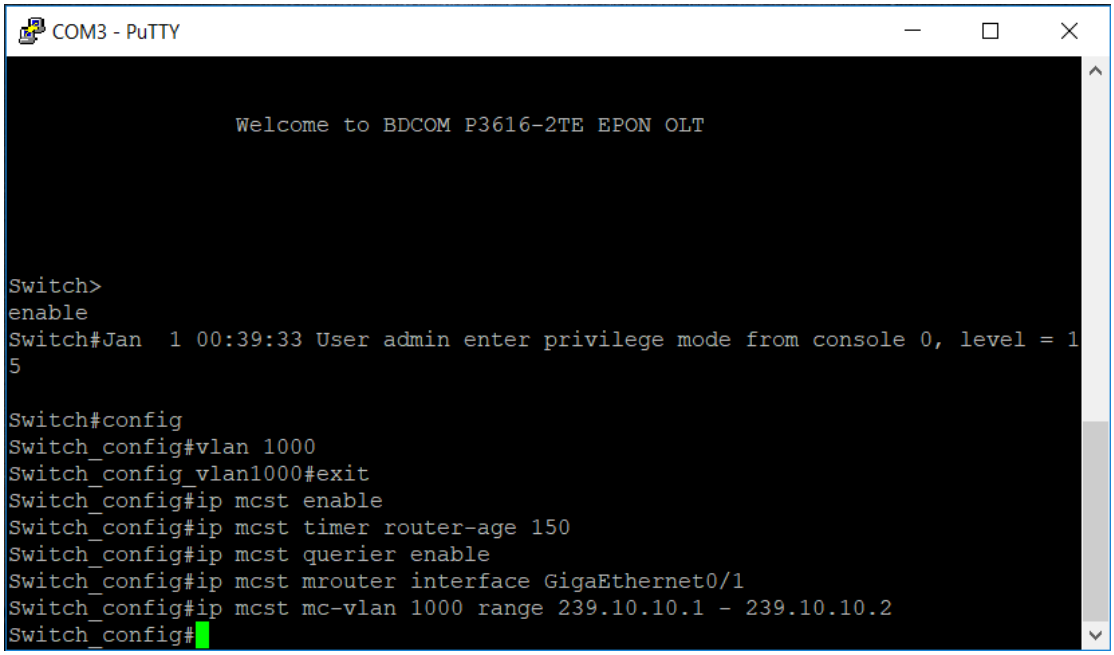
- Switch#config
- Switch_config#vlan 1000
- Switch_config_vlan1000#exit
- Switch_config#



Note: Here the Multicast VLAN is 1000. You may create any VLAN if you want.

To enable Multicast in OLT:

- Switch#config
- Switch_config#ip mcst enable
- Switch_config#ip mcst timer router-age 150
- Switch_config#ip mcst querier enable
- Switch_config#ip mcst mrouter interface GigaEthernet0/1
- Switch_config#ip mcst mc-vlan 1000 range 239.10.10.1 - 239.10.10.2



Note: Here we consider our uplink port is Giga0/1 and the IP range is 239.10.10.1 - 239.10.10.2. You can change it if you want.

To configure the interface (Uplink and PON) in OLT:

- Switch#config
- Switch_config#interface gigaEthernet 0/1
- Switch_config_g0/1#switchport mode access
- Switch_config_g0/1#switchport pvid 1000
- Switch_config_g0/1#exit

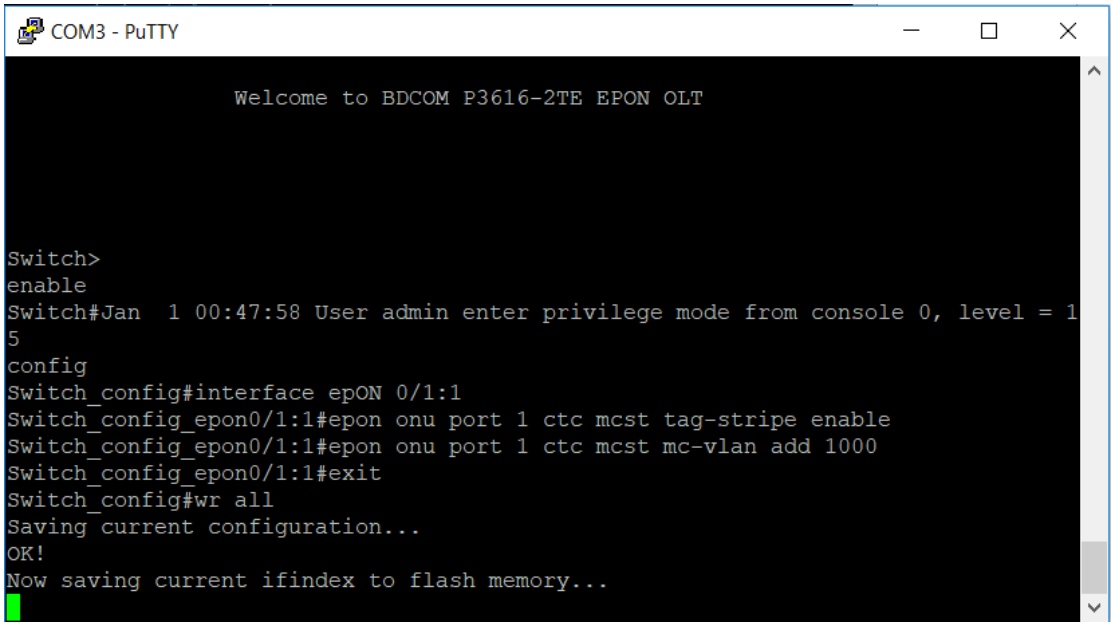
- Switch_config#interface epoN 0/1
- Switch_config_epon0/1#switchport mode trunk
- Switch_config_epon0/1#exit
- Switch_config#write all

```
COM3 - PuTTY

                 Welcome to BDCOM P3616-2TE EPON OLT




Switch>
enable
Switch#Jan  1 00:43:51 User admin enter privilege mode from console 0, level = 1
5

Switch#config
Switch_config#vlan 1000
Switch_config_vlan1000#exit
Switch_config#ip mcst enable
Switch_config#ip mcst timer router-age 150
Switch_config#ip mcst querier enable
Switch_config#ip mcst mrouter interface GigaEthernet0/1
Switch_config#ip mcst mc-vlan 1000 range 239.10.10.1 - 239.10.10.2
Switch_config#interface gigaEthernet 0/1
Switch_config_g0/1#switchport mode access
Switch_config_g0/1#switchport pvid 1000
Switch_config_g0/1#exit
Switch_config#interface epoN 0/1
Switch_config_epon0/1#switchport mode trunk
Switch_config_epon0/1#exit
Switch_config#write all
Saving current configuration...
OK!
Now saving current ifindex to flash memory...
OK!
```

Note: Here we consider our uplink port is Giga0/1 and PON port is epon0/1

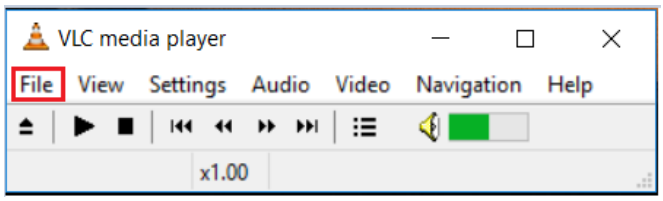
To configure the ONU:

- Switch#config
- Switch_config#interface epON 0/1:1
- Switch_config_epon0/1:1#epon onu port 1 ctc mcst tag-stripe enable
- Switch_config_epon0/1:1#epon onu port 1 ctc mcst mc-vlan add 1000
- Switch_config_epon0/1:1#exit
- Switch_config#write all



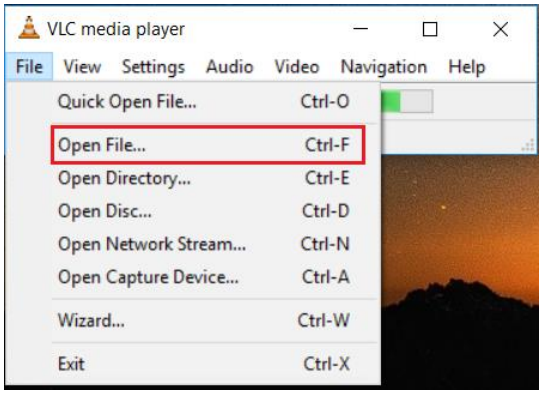
Note: If the server MAC address is private MAC address then you have to use the command ***epon local-mac forward*** in config mode to resolve the issue

Multicast testing procedure:

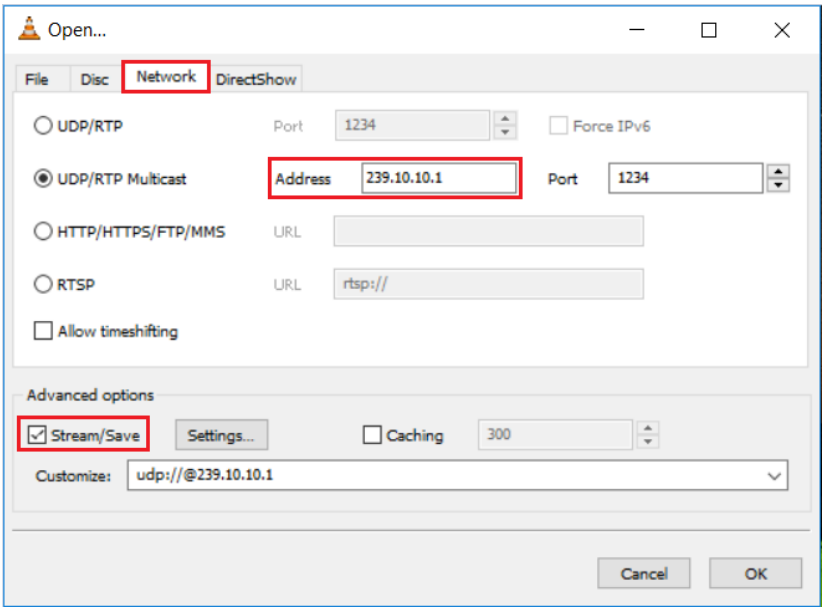


To configure client side:

1. Install VLC media player to your client PC
2. Open VLC media player and click on **File**

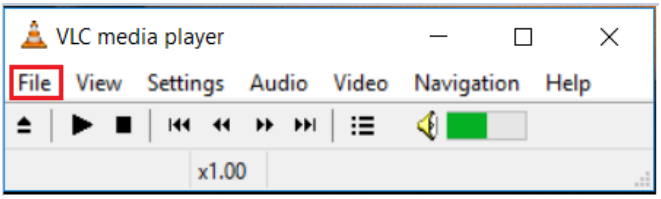


3. Click on **Open File**



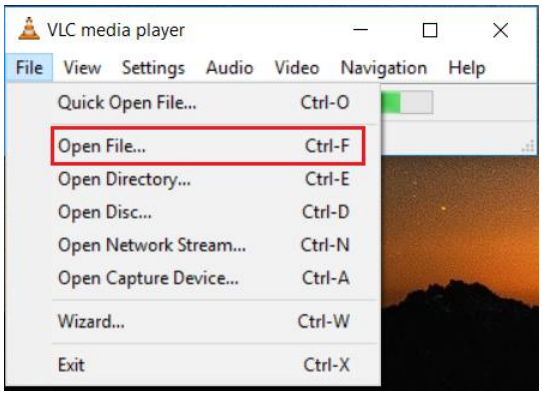
4. Click on **Network**
5. Select UDP/RTP Multicast and configure your server IP Address. Here, we can consider the server IP is 239.10.10.1
6. **Stream/Save** should be tick marked

Note: Here we used VLC media player version 0.8.5

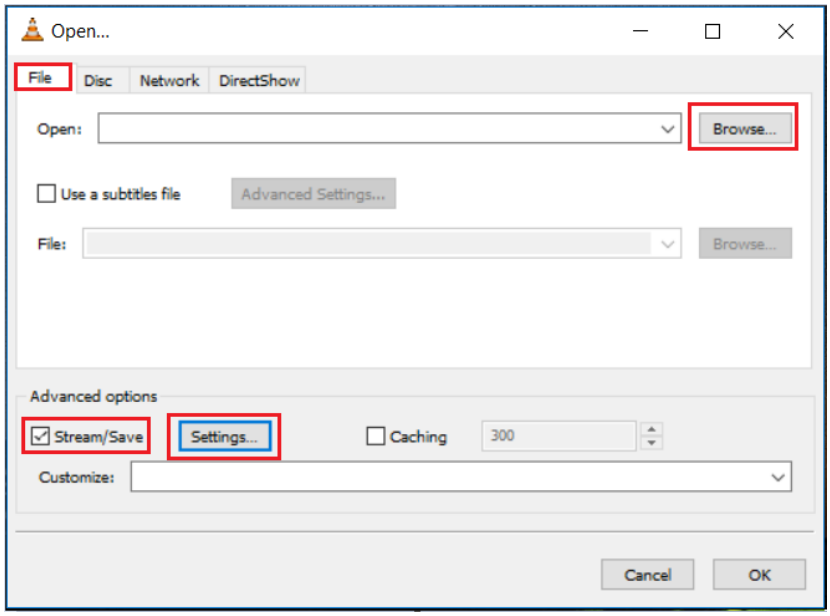


To configure the Server:

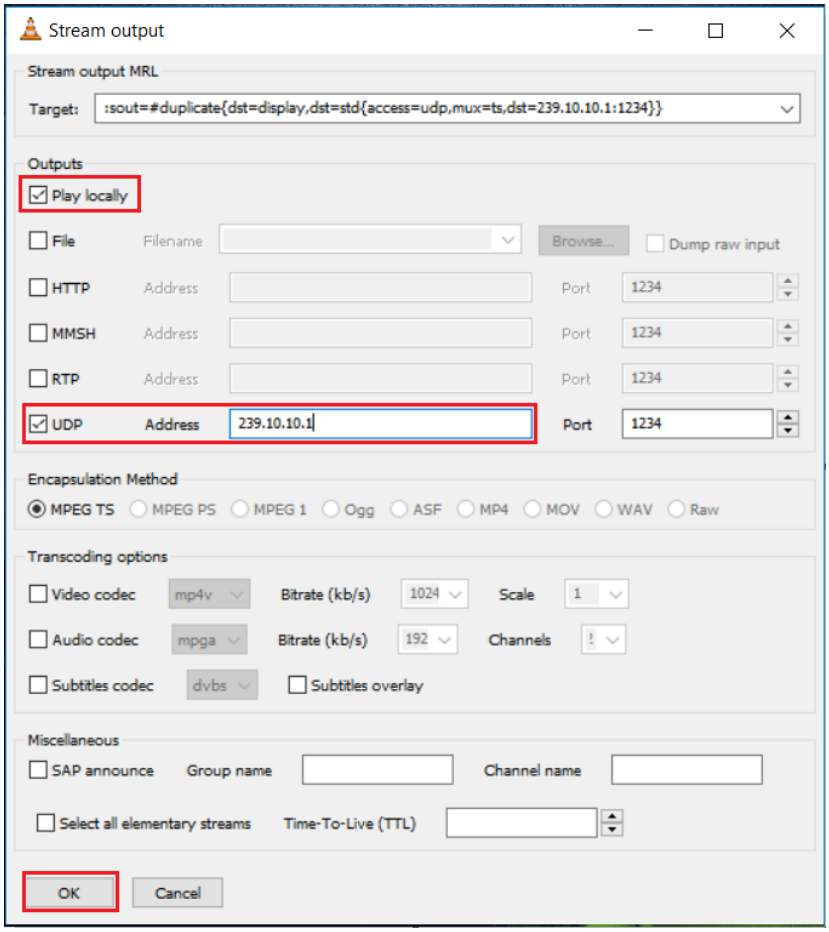
1. Install VLC media player to your client PC
2. Open VLC media player and click on **File**



3. Click on **Open File**

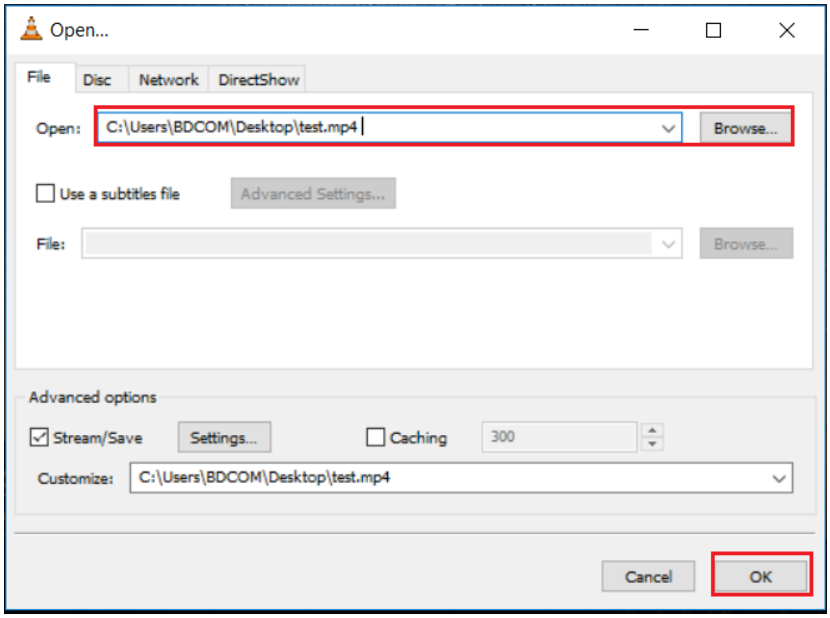


4. Select **File** tab
5. Browse the video file that you want to show to your client
6. **Stream/Save** should be tick marked
7. Click on **Settings**



8. **Play locally** should be tick marked

9. Select **UDP** and in Address filed configure your server address
10. Click on **OK**



11. Select your video file that you would like to show to your client by browsing your file location
12. Click on **OK**

Note: Whenever you have done the configuration on server side the video will be played at your client side.